

■ MCQ Questions: Stack (Without Answers)

1. A data structure that stores elements in a linear order is called
 - (A) Non-linear data structure
 - (B) Hierarchical data structure
 - (C) Linear data structure
 - (D) Distributed data structure
2. Which of the following is **not** a linear data structure?
 - (A) Stack
 - (B) Queue
 - (C) Graph
 - (D) List
3. Stack follows which principle?
 - (A) FIFO
 - (B) LIFO
 - (C) FILO
 - (D) LILO
4. In a stack, insertion and deletion take place at
 - (A) Bottom only
 - (B) Both ends
 - (C) Middle
 - (D) Same end
5. The end of the stack where insertion and deletion occur is called
 - (A) Base
 - (B) Pointer
 - (C) Top
 - (D) Index
6. Which real-life example best represents a stack?
 - (A) Queue at a ticket counter
 - (B) Stack of plates
 - (C) Railway timetable
 - (D) Tree structure
7. Which operation inserts an element into the stack?
 - (A) POP
 - (B) PUSH
 - (C) PEEK
 - (D) DELETE
8. Which operation removes the top element of the stack?
 - (A) PUSH
 - (B) INSERT
 - (C) POP
 - (D) ADD
9. Attempting to insert an element into a full stack causes
 - (A) Underflow

- (B) Overflow
 - (C) Runtime error
 - (D) Index error
10. Attempting to remove an element from an empty stack causes
- (A) Overflow
 - (B) Syntax error
 - (C) Underflow
 - (D) Name error
11. Which Python data type is commonly used to implement stack?
- (A) Tuple
 - (B) Set
 - (C) Dictionary
 - (D) List
12. Which list method is used to PUSH an element in stack implementation?
- (A) insert()
 - (B) extend()
 - (C) append()
 - (D) add()
13. Which list method is used to POP an element from stack?
- (A) remove()
 - (B) delete()
 - (C) pop()
 - (D) clear()
14. In Python stack implementation, explicit declaration of TOP is not needed because
- (A) Python has automatic memory management
 - (B) Lists are unordered
 - (C) append() and pop() work at the end
 - (D) Stack size is fixed
15. Which function is used to check whether a stack is empty?
- (A) size()
 - (B) top()
 - (C) isEmpty()
 - (D) empty()
16. Which operation returns the number of elements in the stack?
- (A) size()
 - (B) count()
 - (C) length()
 - (D) total()
17. Which function returns the topmost element without removing it?
- (A) pop()
 - (B) push()
 - (C) top()
 - (D) display()

18. Which of the following applications uses stack?
- (A) CPU scheduling
 - (B) Undo/Redo operations
 - (C) File indexing
 - (D) Database normalization
19. Web browser BACK button works using
- (A) Queue
 - (B) Stack
 - (C) Tree
 - (D) Graph
20. Matching of parentheses in expressions is done using
- (A) Queue
 - (B) Stack
 - (C) Array
 - (D) Dictionary
21. Which notation places operators between operands?
- (A) Prefix
 - (B) Postfix
 - (C) Infix
 - (D) Polish
22. Which notation is also known as Reverse Polish Notation?
- (A) Prefix
 - (B) Infix
 - (C) Postfix
 - (D) Functional
23. Expression $x y +$ is written in
- (A) Infix notation
 - (B) Prefix notation
 - (C) Postfix notation
 - (D) Binary notation
24. Which notation does **not** require parentheses?
- (A) Infix
 - (B) Prefix
 - (C) Postfix
 - (D) Both B and C
25. Which rule is followed while evaluating infix expressions?
- (A) LIFO
 - (B) FIFO
 - (C) BODMAS
 - (D) FILO
26. During infix to postfix conversion, which elements are pushed onto stack?
- (A) Operands
 - (B) Operators

- (C) Numbers only
 - (D) Variables only
27. During postfix expression evaluation, which elements are pushed onto stack?
- (A) Operators
 - (B) Operands
 - (C) Parentheses
 - (D) Expressions
28. How many operands are popped when an operator is encountered during postfix evaluation?
- (A) One
 - (B) Two
 - (C) Three
 - (D) None
29. In postfix evaluation, the final result is obtained when the stack has
- (A) No elements
 - (B) Two elements
 - (C) One element
 - (D) Multiple elements
30. If stack contains more than one element after postfix evaluation, it indicates
- (A) Correct expression
 - (B) Overflow condition
 - (C) Invalid postfix expression
 - (D) Syntax error
31. Which data structure is best suited for expression conversion?
- (A) Queue
 - (B) Stack
 - (C) Linked list
 - (D) Tree
32. PUSH and POP operations in stack work in
- (A) Random order
 - (B) FIFO order
 - (C) LIFO order
 - (D) Sorted order
33. Which of the following is **not** an application of stack?
- (A) Function calls
 - (B) Parenthesis matching
 - (C) Undo operation
 - (D) Level-order traversal
34. Stack is mainly used for
- (A) Sequential access
 - (B) Hierarchical storage
 - (C) Temporary storage
 - (D) Random access

35. Which of the following correctly represents postfix notation?

- (A) $A + B$
- (B) $+ A B$
- (C) $A B +$
- (D) $(A + B)$

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